MIL-T-12625D(ME)
4 August 1981
SUPERSEDING
MIL-T-12625C(ME)
21 May 1976

MILITARY SPECIFICATION

THERMOMETER, SELF-INDICATING, LIQUID-IN-GLASS, POCKET TYPE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a liquid-in-glass, self-indicating thermometer graduated in Fahrenheit (F) degrees.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.

FSC 6685

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The thermometer shall be etched-stem, liquid-in-glass, total-immersion type, graduated in Fahrenheit degrees, as shown in Figure 1, and as specified herein.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

3.3 Storage. The thermometer shall not be damaged during storage at any ambient temperature from minus 66° F to plus 160° F (minus 54° C to plus 71° C).

3.4 Thermometer tube.

3.4.1 Stem. The stem shall be magnifying type glass, and shall have a broad strip of white glass at the back to increase reflected light. The stem shall be designed to increase the angle of vision and effective width of the thermosensitive fluid column and to accommodate the graduations and numbers on the flat front of the tube. The stem shall be annealed. A button top shall be provided for holding the thermometer securely in the case cap.

3.4.2 Bulb. The bulb shall be normalized thermometer glass, cylindrical in shape, or shall be blown from the stem.

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3.4.3 Thermosensitive fluid. The bulb and tube shall contain a sufficient quantity of fluid so that, when correctly calibrated, the height of the fluid in the tube shall indicate the temperature of the medium surrounding the thermometer within the range and accuracy specified herein. The type fluid used shall minimize any tendency for the liquid column to separate. A non-fading dye may be used to color the fluid to allow a contrast for reading.

3.4.4 Characteristics. The thermometer tube shall conform to the following requirements:

Immersion	- Total
Temperature range	- Minus 66° F to plus 160° F
Graduation (space between)	- 2 Fahrenheit degrees
Width of graduation lines	- 0.004, plus 0.001 minus 0.000 inch
Longer graduation lines at each	- 10° F
Graduations numbered without "°" mark or letter "F" at each multiple of	- 20° F
Scale error at any point when standardized shall be not more than	- 3° F
Standardization	- Ice point and every 20° F also minus 48° F
Special inscription on thermometer	- Letter "F" etched above the upper limit of the scale

3.4.5 Graduations. The graduation marks and numbers shall be etched on the stem, filled with an indelible pigment of distinctive color, and shall be of sufficient size to be clearly legible from a distance of 18 inches. For temperatures of zero degrees and above, the color shall be black. For temperatures below zero degrees, the color shall be red. The pigment shall not be affected by submersion for 10 days in a solution of not less than 75 percent sulfuric acid or a commercial grade or toluene at 100° F (38° C).

3.5 Case. When specified (see 6.2), the thermometer shall be furnished with a case. The case shall be aluminum or brass and shall have an anodic or chemical black finish, as applicable. The case shall be 0.44 inch outside diameter, plus or minus 0.06 inch, and shall have a wall thickness of not less than 0.03 inch. The top shall be a two-piece, knurled, internal-external screw cap which will engage the button on the top of the stem. The cap shall be equipped with a protective guard and shall hold the thermometer centered within the case. The cap shall have a lanyard ring at the top. A ball-pointed pocket clip shall be affixed near the upper end of the case in such a way so that it will not slide along the case. The overall length of the case shall be not less than 6.0 inches and not more than 6.5 inches.

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3.6 Identification marking. Each thermometer and its case (when a case is provided) shall be identified in accordance with MIL-STD-130.

3.7 Workmanship. The thermometer and thermosensitive fluid shall be free from gas bubbles and foreign material. The thermometer shall be clear and free from pits, chips, sharp edges, or other defects which would prove detrimental to accuracy and appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for the minor defect.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective.

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4.4 Inspection procedure.

4.4.1 Examination. The thermometers shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Stem not as specified.
103. Temperature range not as specified.
104. Graduation not as specified.
105. Separation or bubbles in the thermosensitive fluid.
106. Case not as specified (when a case if furnished).
107. Workmanship not as specified.

Minor

201. Identification marking on the thermometer or case (when a case is furnished) missing, incomplete or illegible.

4.4.2 Tests.

4.4.2.1 Test conditions. A thermometer which has been certified by the National Bureau of Standards to have an accuracy of plus or minus 0.1° F shall be used in the test specified in 4.4.2.3.

4.4.2.2 Storage. Sample thermometers shall be placed in a cold chamber at a temperature of minus 66° F (minus 54° C) for not less than 3 days without solar radiation. The thermometers shall then be removed, allowed to return to room temperature, and placed in an oven at plus 160° F (plus 71° C) for 4 hours. The thermometers shall be removed from the oven, allowed to return to room temperature, and examined. Any evidence of breaks, cracks, or separation bubbles in the thermosensitive fluid shall constitute failure of this test.

4.4.2.3 Standardization. The certified thermometer and the sample thermometers shall be totally immersed in a well-stirred water bath and tested at the standardization points specified in 3.4.4. Failure of the sample thermometers to indicate the same reading as the certified thermometer within 3° F shall constitute failure of this test.

4.4.2.4 Marking. Each of the sample thermometers shall be submerged in a solution of not less than 75 percent sulfuric acid or a commercial grade of toluene at plus 100° F (plus 38° C) for a period of 10 days. At the end of the test period, the sample thermometer shall be examined. Any damage or diminution of the markings shall constitute failure of this test.

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4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Materials, methods, or containers not as specified.
109.	Thermometers without a case not cushioned as specified.
110.	Consolidation not as specified.
111.	Quantities packed together exceed specified quantities for Level A or Level B.
112.	Marking illegible, incomplete, incorrect, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Thermometer with case. Each thermometer in its case shall be preserved in accordance with MIL-P-116, Method IC-3.

5.1.1.2 Thermometer without case. Each thermometer shall be cushioned in a fiber or plastic tube or paperboard box and shall then be preserved in accordance with MIL-P-116, Method IC-3.

5.1.1.3 Consolidation. Thermometers of like description shall be consolidated together in quantities not to exceed 20 pounds per box, in close-fitting boxes conforming to PPP-B-636, W6c, style optional.

5.1.2 Commercial. Thermometers shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

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5.2.1 Level A. Thermometers preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional in quantities not to exceed 200 pounds. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Thermometers, preserved as specified in 5.1, shall be packed together in quantities not to exceed the size or weight limitation of the box, in close-fitting boxes conforming to PPP-B-636, V3c, style optional. Box closure and sealing shall be as specified for Method V in the appendix to the box specification.

5.2.3 Commercial. Thermometers shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The thermometers are intended for use in determining atmospheric temperatures during surveying operations and for use in sling psychrometers of surveying altimeters contained in MIL-A-13170.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a case is required (see 3.5).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

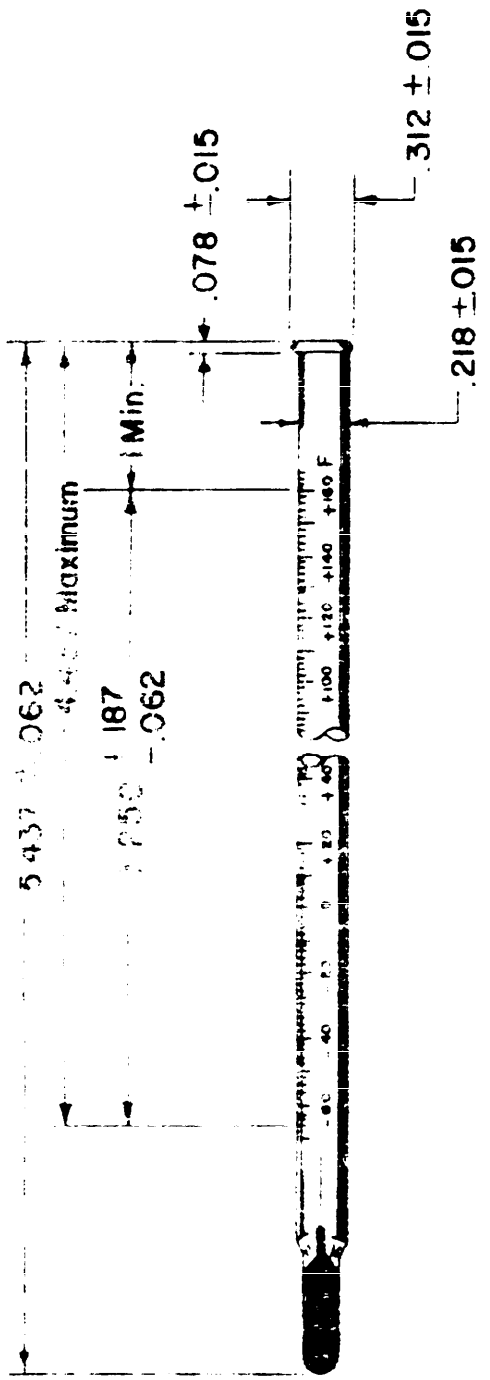
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NOTE:
DIMENSIONS ARE IN INCHES.

FIGURE 1. Thermometer tube.

X-3130A

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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-T-12625D(ME) Thermometer, Self-Indicating, Liquid-In-Glass, Pocket Type

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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